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lululemon Labs Standard Operating Procedures: Breast Movement Data Collection with IMUs



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1 Introduction:

The Vicon IMeasureU sensors (IMU) were designed to capture body segment accelerations during a wide range of activities, both in lab and in field. This guide will review how to collect “breast motion data” for the purposes of validating the UBCO breast torso.



2 Test protocol summary

2.1.1 Test protocol requirements

- Blue Trident IMUs (x3)
- Blue Trident USB Docks (x3)
- USB 10-port charger
- IMU adhesives/double sided tape
- Collection device (iPad with Capture U app)
- Bra samples

2.1.2 Test outputs for analysis

- One highG .csv file for each sensor, for each trial
- One 'Aligned' .csv file for each sensor, for each trial
 - Has an elapsed time timestamp from when all sensors started collecting

3 Collection Protocols:

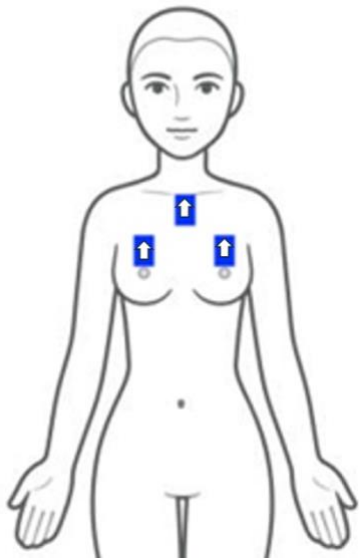
3.1 Collecting Data



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3.1.1 IMU Placement

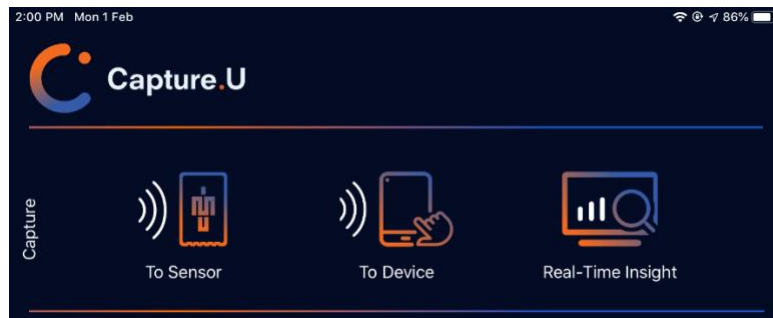
1. Place a double-sided adhesive on the back of the sensor. Remove the backing and affix it to the torso in the locations outlined in the below table. All sensors are numerically labelled with their ID, with an arrow indicating +ve X direction. Typically, the +ve X arrow points towards the proximal end of the segment, along the long axis.
2. **Create a spreadsheet in your data collection folder and record which IMU number was placed on which body location.** (Helpful for post processing if they are placed in ascending order)

IMU	Position	Arrow Direction	
Sternum	On the centre of the sternum, below the centre of the collar bones	Arrow pointing up	
R Breast	On the right breast, just above the nipple with the arrow pointing up.	Arrow pointing up	
L Breast	On the left breast, just above the nipple.	Arrow pointing up	

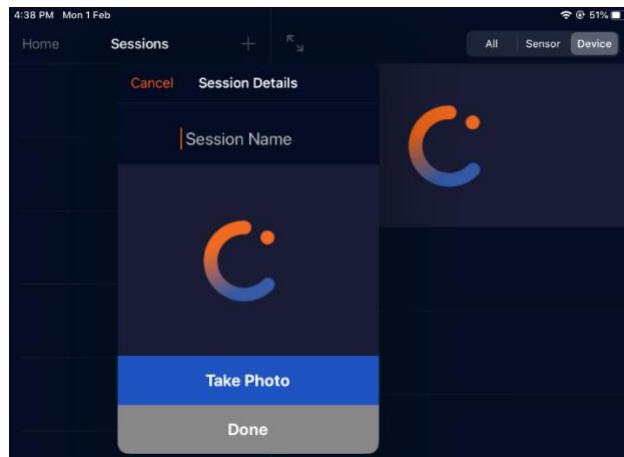
3.1.2 Collect a trial

1. In the Capture.U iOS app, select “To Sensor”

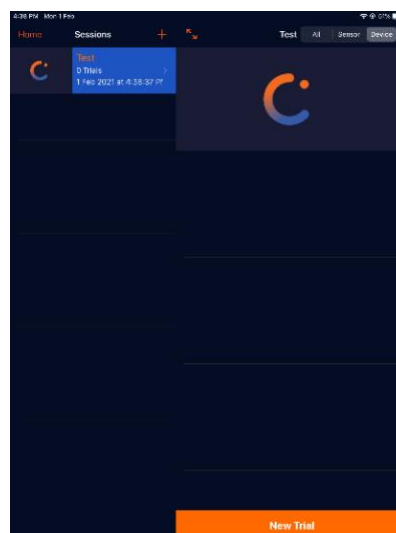
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2. In the top left, click the “+” button to create a new session. Name the session “UBCO Bra Rig Tests”



3. Select the session you just created. The session box will highlight blue and the new trial button at the bottom will turn orange.

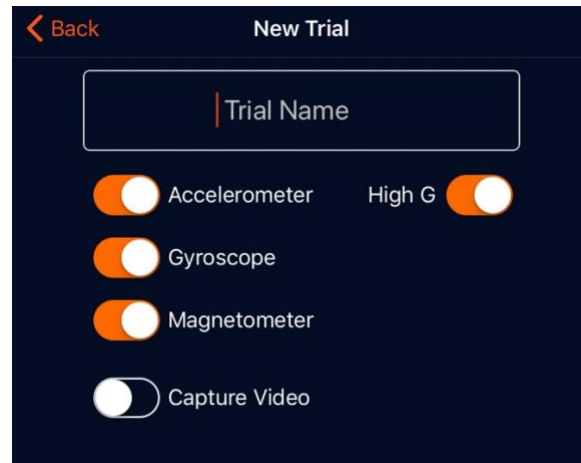


4. Click new trial and type in the trial/condition name with the below naming convention.

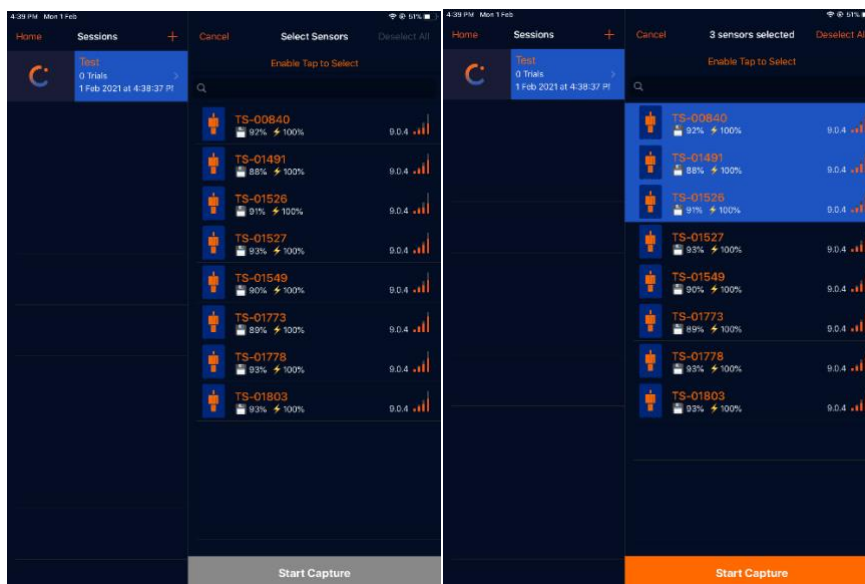


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- a. BraRigTest-BraRig-BraSize-BraName1
 - i. Replace BraSize with the toro's size
 - ii. Replace BraName with the name of the bra
 - iii. If more than one trial in each bra is being performed, add a number after the trial name, but ensure there is **nothing** separating the number from the bra name
5. Ensure that accelerometer, gyroscope, magnetometer and High G are turned on.



6. Click “Start Capture” (capture doesn’t actually start yet). Select the sensors you are using from the list of all detected sensors.

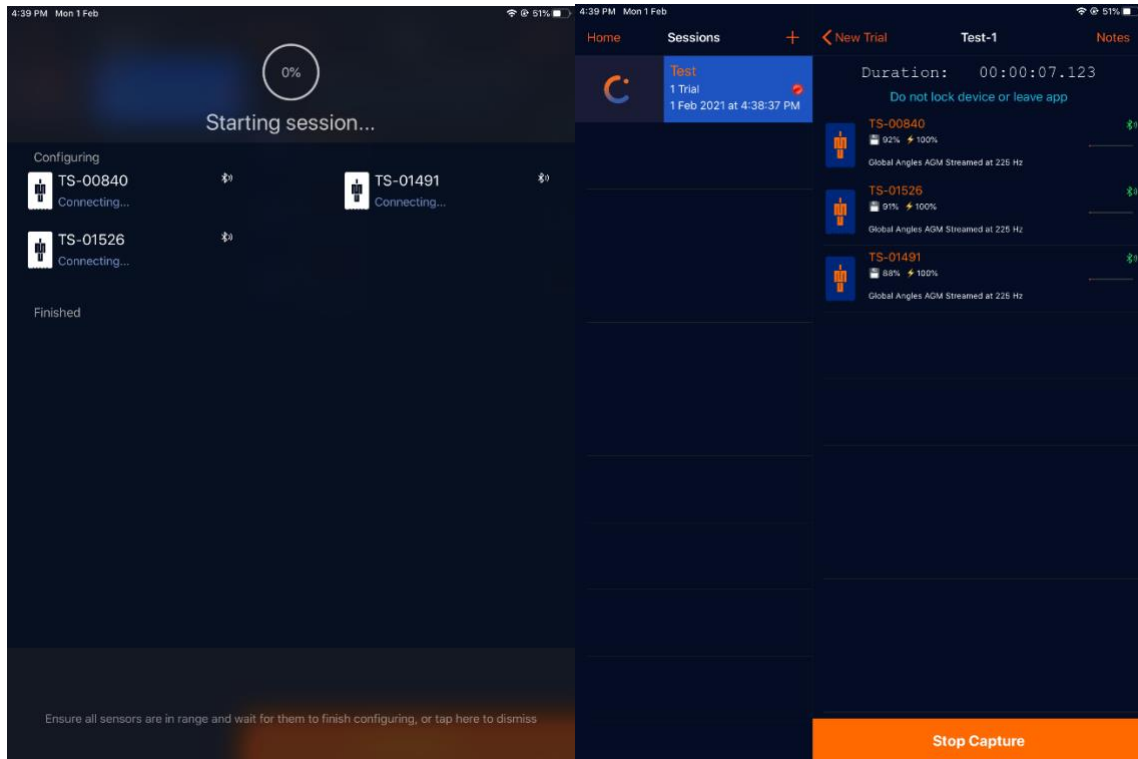


7. Click “Start Capture” (capture doesn’t actually start again). After you click this button, the sensors configure and synchronize and will start recording at different times. After



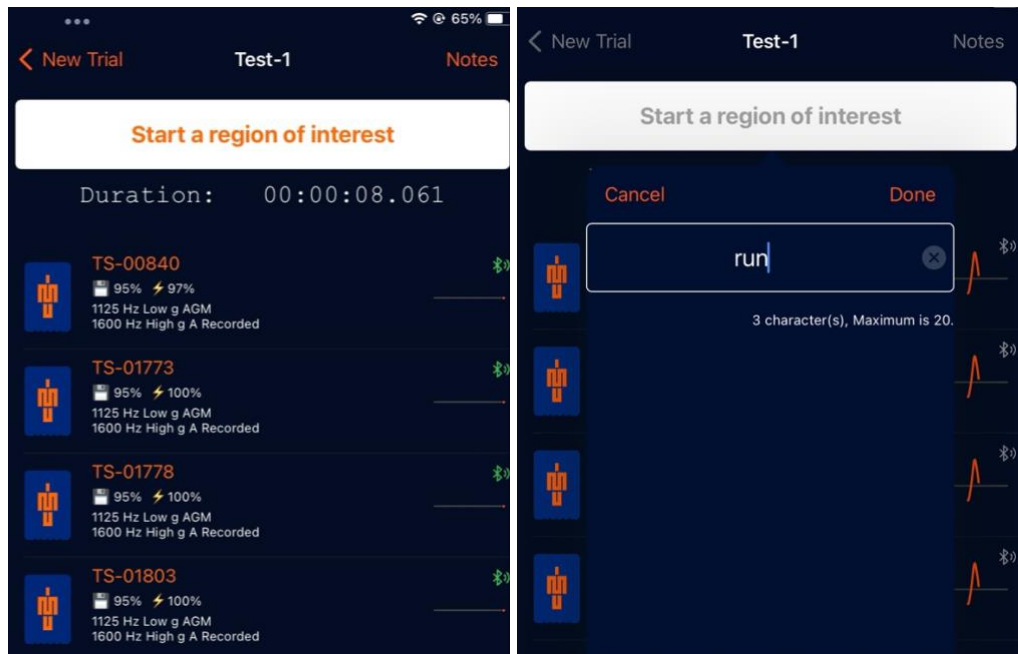
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the starting session screen disappears, the main screen will show a duration clock at the top right and a pulse wave beside the sensor names indicating recording is happening. The “duration” label indicates the time it took for all sensors to become synchronized and begin collecting data.

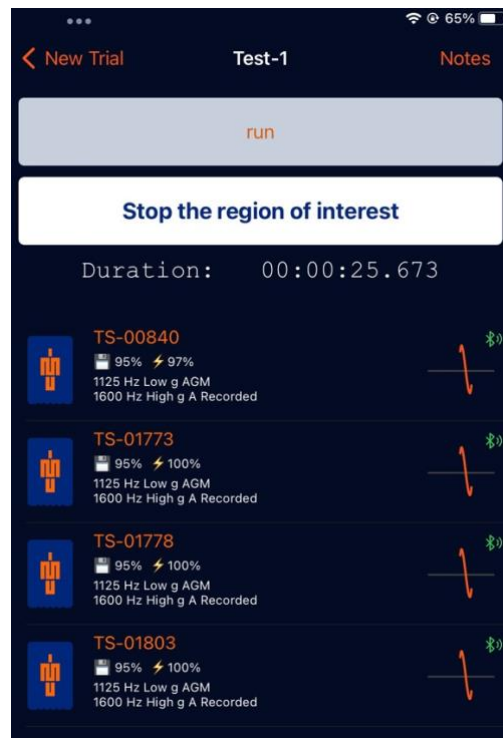


- a. NOTE: While saving to the sensor, the sensors may go out of range of the iPad. The sensors must be in range to start or stop a trial.
8. Once the sensors are recording, select “Start a region of interest,” then select “run”. If “run” does not appear, select “add new region name” and name this region “run.” Ensure a few seconds (3-5s) of quiet standing are recorded, then turn the bra rig on. This period of quiet standing is required within the region of interest for accurate data analysis.

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9. Once you are finished with the trial, click “Stop region of interest” then click “Stop Capture.” After a brief stopping session screen, you will be returned to the session screen with your new trial added to the list.



10. To collect a new trial, repeat steps 4-9.

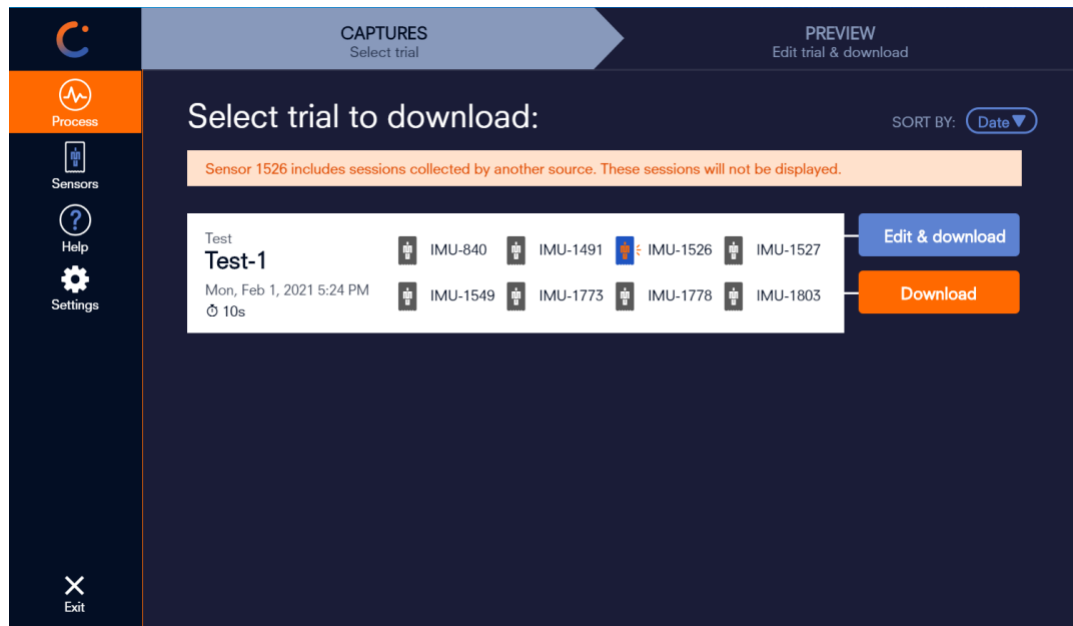


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3.1.3 Exporting Data

Because the data is being saved to the sensors, they must be connected to a computer for download.

- Place the IMUs that you wish to download into the dock and use the USB cord to connect the dock to the computer. Ensure CaptureU is downloaded onto the computer.
- Once the sensors are plugged in, the Process tab appears in the Capture.U software. This tab shows all the trials saved on the sensors, and which sensors collected that trial. The meaning of each sensor icon is shown below.

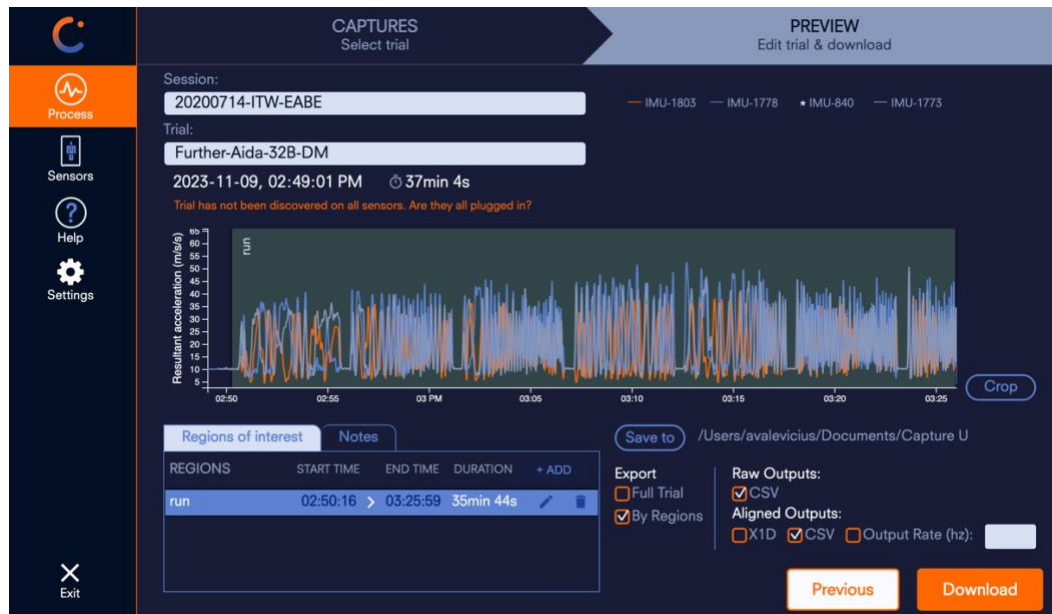


Icon	Meaning
 IMU-186	Blue Trident sensor is connected to Capture.U Desktop.
 IMU-4	Blue Trident sensor is not connected to Capture.U Desktop.
 IMU-3	Blue Trident sensor was selected for the trial, but no data has been recorded or data has been erased.
 IMU-31	Blue Trident sensor data has been downloaded for the trial.

- Select “Edit & Download,” you will be brought to a data preview screen.
 - a. The acceleration data is plotted in the preview window. The shaded region indicates the area of interest. If required, edit this area of interest so that it begins with the quiet standing (flat line). This can be edited by clicking the plot and dragging the edges of the shaded region.
 - b. Select the folder you would like to save the data to.

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- c. Ensure “Export by Region,” is selected and that Aligned outputs CVS is checked.
- d. Click “Download.”



3.2 Bras

The following bras have been sent to UBCO, one of each size listed:

Bra Name	Sizes	File naming convention
No bra	34	TEST-BRARIG-34D-NOBRA001
No bra	36	TEST-BRARIG-36C -NOBRA001
Energy Medium Support	8 (34)	TEST-BRARIG-34D-ENERGY001
Energy Medium Support	10 (36)	TEST-BRARIG-36C-ENERGY001
Like a Cloud	8 (34)	TEST-BRARIG-34D-LIKEACLOUD001
Like a Cloud	10 (36)	TEST-BRARIG-36C -LIKEACLOUD001



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FlowY	8 (34)	TEST-BRARIG-34D- FLOWY001
FlowY	10 (36)	TEST-BRARIG-36C - FLOWY001
Energy High Support	34D	TEST-BRARIG-34D- ENERGYHIGH001
Energy High Support	36C	TEST-BRARIG-36C - ENERGYHIGH001
Ultralu Square Neck Workout Bra	8 (34)	TEST-BRARIG-34D- ULTRALU001
Ultralu Square Neck Workout Bra	10 (36)	TEST-BRARIG-36C - ULTRALU001

Notes

- 001, 002, 003 represent repeated trials after taking off and putting on the bra.
- If different breast tissue is used (e.g. softer breasts), please add on extra conditions for them as a comparison.

